Hands-on Activity 9.1 Tensorflow Exercises

Import

```
import tensorflow as tf
print(tf.__version__)
```

```
2.21.0
```

1. Create a vector, scalar, matrix and tensor with values of your choosing using tf.constant().

```
scalar = tf.constant(7)
vector = tf.constant([6,7])
matrix = tf.constant([[10,7],[6,7]])
tensor = tf.constant([[[1,2,3],[4,5,6]],[[5,6,7],[4,3,1]]])
```

2. Find the shape, rank and size of the tensors you created in 1.

```
print("Shapes")
print(tf.shape(scalar))
print(tf.shape(vector))
print(tf.shape(matrix))
print(tf.shape(tensor))
```

```
Shapes
tf.Tensor([], shape=(0,), dtype=int32)
tf.Tensor([2], shape=(1,), dtype=int32)
tf.Tensor([2 2], shape=(2,), dtype=int32)
tf.Tensor([2 2 3], shape=(3,), dtype=int32)
```

```
print("Ranks")
print(tf.rank(scalar))
print(tf.rank(vector))
print(tf.rank(matrix))
print(tf.rank(tensor))
```

```
Ranks
tf.Tensor(0, shape=(), dtype=int32)
tf.Tensor(1, shape=(), dtype=int32)
tf.Tensor(2, shape=(), dtype=int32)
tf.Tensor(3, shape=(), dtype=int32)
```

```
print("Size")
print(tf.size(scalar))
print(tf.size(vector))
print(tf.size(matrix))
print(tf.size(tensor))
```

```
Size
tf.Tensor(1, shape=(), dtype=int32)
tf.Tensor(2, shape=(), dtype=int32)
tf.Tensor(4, shape=(), dtype=int32)
tf.Tensor(12, shape=(), dtype=int32)
```

3. Create two tensors containing random values between 0 and 1 with shape [5, 300].

```
X = tf.random.uniform(shape=[5, 300], minval=0, maxval=1,
dtype=tf.float32)
Y = tf.random.uniform(shape=[5, 300], minval=0, maxval=1,
dtype=tf.float32)

print(tf.shape(X))
print(tf.shape(Y))

tf.Tensor([  5 300], shape=(2,), dtype=int32)
tf.Tensor([  5 300], shape=(2,), dtype=int32)
```

4. Multiply the two tensors you created in 3 using matrix multiplication.

```
matrix_result = tf.math.multiply(X, Y)
matrix_result.shape

TensorShape([5, 300])
```

5. Multiply the two tensors you created in 3 using dot product.

```
dot_result = tf.tensordot(X, tf.transpose(Y), axes=1)
dot_result.shape

TensorShape([5, 5])
```

6. Create a tensor with random values between 0 and 1 with shape [224, 224, 3].

```
A = tf.random.uniform(shape=[224, 224, 3], minval=0, maxval=1)
```

7. Find the min and max values of the tensor you created in 6.

```
A_max = tf.reduce_max(A).numpy()
A_min = tf.reduce_min(A).numpy()

print(f"Max Value in A: {A_max}")
print(f"Min Value in A: {A_min}")
```

```
Max Value in A: 0.999991774559021
Min Value in A: 1.0251998901367188e-05
```

8. Created a tensor with random values of shape [1, 224, 224, 3] then squeeze it to change the shape to [224, 224, 3].

```
B1 = tf.random.uniform(shape=[1, 224, 224, 3])
B2 = tf.reshape(B1, [224, 224, 3])
print(B1.shape)
print(B2.shape)
```

```
(1, 224, 224, 3)
(224, 224, 3)
```

9. Create a tensor with shape [10] using your own choice of values, then find the index which has the maximum value.

```
c = [7, 6, 24, 5, 11, 14, 6, 100, 101, 102]
C = tf.constant(c)
C.shape
```

```
TensorShape([10])
```

```
C_max_index = tf.argmax(C)
print(f"Max value index: {C_max_index.numpy()}")
```

```
Max value index: 9
```

10. One-hot encode the tensor you created in 9.

```
depth = 3
C_one_hot = tf.one_hot(C, depth)
C_one_hot
```

```
<tf.Tensor: shape=(10, 3), dtype=float32, numpy=
array([[0., 0., 0.],
       [0., 0., 0.],
       [0., 0., 0.],
       [0., 0., 0.],
       [0., 0., 0.],
       [0., 0., 0.],
       [0., 0., 0.],
       [0., 0., 0.],
       [0., 0., 0.],
       [0., 0., 0.]], dtype=float32)>
```

Summarize your learnings in this activity.

The activity provides an introduction on the utilization of Tensorflow and its different capabilities. It demonstrated how different data structures, such as vectors, scalar, matrices and

tensors, are created, used, and manipulated. It also demonstrated how various operations can be applied into different values using Tensorflow. In the